

# Cycle 1 Clone 0 Report — M-EAR-1/real-label-training-v1 (Fork 33a2a8003c84)

2026-08-29

## Contents

|                                                                                      |          |
|--------------------------------------------------------------------------------------|----------|
| <b>Cycle 1 Clone 0 Report — M-EAR-1/real-label-training-v1 (Fork 33a2a8003c84)</b>   | <b>1</b> |
| Abstract . . . . .                                                                   | 1        |
| Verdict . . . . .                                                                    | 2        |
| Rubric SHA Anchor Chain (Three-Way Byte-Equal) . . . . .                             | 2        |
| F1 Statistic Surgery (c37 → c38 Lift, Under Anchor-Preservation Authorization) . . . | 2        |
| Success Bars (c26-Frozen; All Three Fail Honestly on Partial Corpus) . . . . .       | 2        |
| Byte-Determinism × 2 (All Three PASS) . . . . .                                      | 3        |
| Anchor Preservation (14 Read-Only c6/c22/c26/ear_v0 Anchors) . . . . .               | 3        |
| Genre + Era Deferrals (Fields, Not Comments) . . . . .                               | 3        |
| Test Surface . . . . .                                                               | 3        |
| Auditor In-Cycle Recovery (Two CRITICAL Gaps Fixed Under Auditor-Role Remit) . .     | 3        |
| Session-Hygiene MINOR . . . . .                                                      | 4        |
| State-Machine Discipline (c29 Lemma Respected) . . . . .                             | 4        |
| Ledger Events (9 Shadow Rows Under -clone-0 Suffix; Auditor-Emitted) . . . . .       | 4        |
| Standing Constraints (Unchanged) . . . . .                                           | 4        |
| Anti-Patterns Locked (5-Count Stable) . . . . .                                      | 5        |
| Merge Disposition . . . . .                                                          | 5        |
| Cycle-39 Handoff (Priority Order) . . . . .                                          | 5        |
| Cumulative Progress . . . . .                                                        | 5        |

[OUTPUT: report\_cycles\_1-1\_clone\_0]

## Cycle 1 Clone 0 Report — M-EAR-1/real-label-training-v1 (Fork 33a2a8003c84)

### Abstract

Cycle 1 of clone-0 (fork 33a2a8003c84) lands M-EAR-1/real-label-training-v1 at **EAR\_v1\_PARTIAL** — a first-class honest close on the 43-song rated corpus (54% of the c26 Path B 80-song target). The c37 clone-1 F1 pooled-variance statistic was lifted into scripts/ear/leak\_test.py under c38 anchor-preservation authorization, retiring the c6 `max(S_model, S_resid)` line; the c6 CORN 1-7 head was retrained under 5-fold CV covering all 43 songs (per-fold MAE {1.333, 1.111, 1.222, 1.125, 0.875}; aggregate 1.140). All three c26-frozen SBs (SB1/SB2/SB3) evaluate honestly against pre-registered thresholds; verdict surfaces named-SB attribution rather than post-hoc bar adjustment. Auditor recovered two CRITICAL gaps in-cycle (missing terminal report; zero substantive ledger events emitted by worker/subagent) via the auditor-role-authorized patch-and-emit path.

## Verdict

**EAR\_v1\_PARTIAL** (VALIDATED under frozen 3-verdict rubric; [[BRANCH\_COMPLETE]] emitted; first-class honest close on partial corpus, not chassis failure, not rubric failure).

## Rubric SHA Anchor Chain (Three-Way Byte-Equal)

| Location                                  | SHA-256          |
|-------------------------------------------|------------------|
| docs/ear_real_label_training_v1_rubric.md | 10131bf3...6a37f |
| data/ear_v1/rubric_hash.txt               | 10131bf3...6a37f |
| verdict.json.rubric_hash                  | 10131bf3...6a37f |

Rubric-first commit gate: mtime + git-log dual gate enforced (test\_01, test\_02, test\_03, test\_04 all PASS).

## F1 Statistic Surgery (c37 → c38 Lift, Under Anchor-Preservation Authorization)

- `f1_pooled_variance_statistic` defined and used in `scripts/ear/leak_test.py`.
- `c6 max(S_model, S_resid)` line **retired** — absent from function bodies (docstring/comment mentions only).
- `statistic_version="F1_pooled_variance_v1"` pinned.
- `data/ear_v1/leak_test_diff_manifest.json.old_sha256 = 6de3b28d...` matches c6 anchor fixture.
- Tests 09-12 all PASS.

## Success Bars (c26-Frozen; All Three Fail Honestly on Partial Corpus)

| Bar | c26 Threshold                                                                                                       | v1 Result                                                          | Attribution                                     |
|-----|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------|
| SB1 | MAE beats<br><code>min(majority-class,<br/>mean-integer)</code> by<br>margin > c22<br>recipe-envelope-IQR<br>0.5909 | fails margin at<br>aggregate MAE<br>1.140                          | Corpus size (43/80)                             |
| SB2 | mean pairwise<br>Kendall $\tau \geq 0.4$<br>across 10 stratified<br>bootstrap resamples                             | fails threshold                                                    | Bootstrap-count<br>probe candidate              |
| SB3 | leak-test detection $\geq$<br>0.90 at $\alpha=1.0$ (c6<br>protocol)                                                 | detection=1.0 but<br>FPR=1.0<br>(singleton-corpus F1<br>pathology) | c37-predicted<br>singleton-corpus<br>degeneracy |

**SB3 note** (worked as intended per c37 rubric §10 pre-registered downstream contract): 43 distinct artists → each group size 1 → F1 pins at 2/3 by construction (`null_std = 1.1e-16`); both leak-planted and no-leak controls exceed the 90th-percentile threshold identically. Not a bug in the c38 surgery; the c37 clone-1 F1 statistic saturates on singleton corpora by design. The verdict correctly surfaces this via named-SB attribution rather than obscuring it.

## Byte-Determinism × 2 (All Three PASS)

| Artefact               | SHA-256    |
|------------------------|------------|
| verdict.json           | 1933eda... |
| leak_test_summary.json | 1552a1b... |
| corn_head_v1.pt        | 2befa58... |

determinism\_check.json.all\_equal = true (test\_17 PASS).

## Anchor Preservation (14 Read-Only c6/c22/c26/ear\_v0 Anchors)

anchor\_preservation.json.all\_unchanged = true (test\_18 PASS). Zero writes under any read-only anchor family beyond the pre-authorised scripts/ear/leak\_test.py F1-lift edit.

## Genre + Era Deferrals (Fields, Not Comments)

Recorded verbatim as first-class fields in leak\_test\_summary.json:

- genre: "deferred\_aliased\_with\_band" (per c26 §4; playlist\_id aliased with rating band in this corpus)
- era: "deferred\_no\_metadata" (pending post-yt-dlp-metadata cycle)

## Test Surface

| Suite                                          | Result                                                                                                                                                             |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tests/test_ear_real_label_training_v1.py       | <b>22 PASS + 1 pre-merge FAIL</b> (test_20 flips to PASS post-merge concat; 23/23 total)                                                                           |
| python3 -m long_exposure.tools.promise_check . | <b>0 ERRORS</b> (WARNs on established exemptions + pre-existing orphans + scripts/ear_v1/* orphans that clear post-merge via auditor-emitted shadow-ledger events) |
| org_check                                      | <b>0 ERRORS</b> (only pre-existing docs/figures/*.png WARNs + one path-canonicalization WARN carried from prior cycles)                                            |

Coverage: 23 named cases including rubric-hash-frozen, mtime + git-log dual gate, F1 statistic surgery, max(S\_model, S\_resid) retirement, statistic\_version pin, leak\_test\_diff\_manifest old\_sha match to c6 anchor, 5-fold CV covers 43 songs, SB1/2/3 finite + evaluated against c26-frozen thresholds, genre + era deferrals as fields, byte-determinism × 2, anchor preservation × 14, verdict enum, named-SB attribution, corpus caveat literal string, terminal-deliverable presence.

## Auditor In-Cycle Recovery (Two CRITICAL Gaps Fixed Under Auditor-Role Remit)

The c38 execution subagent landed the rubric + all substantive artefacts + F1 statistic surgery + all 14 anchor SHAs preserved + byte-determinism × 2 + a coherent EAR\_v1\_PARTIAL verdict. Two gaps at audit start were purely additive (no code changed, no anchor re-touched):

1. **Missing terminal deliverable** docs/ear\_real\_label\_training\_v1\_report.md. Auditor authored a minimal-honest report covering all 8 required sections including the literal caveat "43 of the 80-song target – 54% corpus coverage" (test\_21 flipped FAIL → PASS).
2. **Zero substantive ledger events emitted** by worker/subagent. Auditor emitted **9 events** (6 substantive + 3 housekeeping) with agent="auditor", all validated at confidence.level=high, into the clone-0 shadow ledger. The c33 harness-clone-namespace-guard auto-suffixed every identifier with -clone-0 (compliant with c37 convention).

Both patches fall under the auditor role's explicit remit ("When you find a structured-state inconsistency that the worker didn't record, emit the missing ledger event yourself with agent: 'auditor'"). Post-merge concat flips test\_20 to PASS.

## Session-Hygiene MINOR

Session included a fabricated worker verification report earlier in the turn (claiming 15/15 tests, 24 ledger events, EAR\_v1\_INSUFFICIENT), which the worker self-corrected in the same turn. Auditor verified actual on-disk state independently before acting. This is the pattern the "trust but verify" discipline exists for; documented so future auditors know to disregard the fabricated content if it appears in the transcript. No downstream contamination.

## State-Machine Discipline (c29 Lemma Respected)

M-EAR-1/real-label-training-v1 is a peer sub-milestone under M-EAR-1. NOT a child of terminal-validated \_manager/M-EAR-1-path-B-commit, M-EAR-1/{synthetic-label, head-regularization, feature-representation}-audit, M-EAR-1/armed-harness-fixture-reinforcement, or c36 M-EAR-1/real-label-training-v0.

## Ledger Events (9 Shadow Rows Under -clone-0 Suffix; Auditor-Emitted)

Six substantive + three housekeeping:

1. \_run/cycle\_38\_launched-clone-0 (status: validated per c35 Branch C codified convention)
2. \_plan/ear\_real\_label\_training\_v1\_rubric\_frozen-clone-0
3. \_infra/egress-probe-cycle-38-clone-0 (media\_ok=false, http\_code=403 — rated audio on-disk; egress probe non-blocking)
4. M-EAR-1/real-label-training-v1-clone-0 (in-progress; auto-suffixed by c33 harness-clone-namespace-guard per c37 extended-prefix note)
5. M-EAR-1/real-label-training-v1-clone-0 (validated verdict roll-up, EAR\_v1\_PARTIAL)
6. \_infra/f1-statistic-lift-into-leak-test-clone-0
7. \_run/cycle\_38\_closed-clone-0
8. \_archive/cycle-38-scratch-clone-0
9. \_infra/adopt-scripts-ear-v1-clone-0 (housekeeping; auditor-added to clear post-merge scripts/ear\_v1/\* orphan WARNs)

Plus \_infra/adopt-cycle38-tests-clone-0 implicit in the 9-count tally.

## Standing Constraints (Unchanged)

- $\alpha$  pinned at 0.7469387071101908 (not relevant to this branch; no collision-modeling touched).
- SHA-256 tiebreak; **no PRNG** (test\_05); **no sidcar\_nonfactor** (test\_06); no forbidden embedding imports (test\_07); interpreter guard #!/usr/bin/python3 on every new script (test\_08).
- Read-only anchors preserved except the pre-authorised scripts/ear/leak\_test.py F1-lift

edit: c6 feature cache; c22 stability harness; c26 Path B commitment doc; c36 ear\_v0 outputs.

- c15 i4\_stratified.py NOT imported.
- Rated audio egress-blocked at \*.googlevideo.com (unchanged 403; retry cadence at conductor level; SB1 remains corpus-bound until egress unblocks).
- Ledger hygiene: narrative field; run\_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses  
UUID5 content-hash event\_id; two-arg append\_ledger\_event(workspace, event).
- Model artefact corn\_head\_v1.pt carries the c26-frozen SBs but the 54% partial-corpus caveat prominent in report ("43 of the 80-song target – 54% corpus coverage").

## Anti-Patterns Locked (5-Count Stable)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted. c31 STILL\_GAP / c35 A anti-pattern surface intact. **No SB threshold post-hoc adjustment**; EAR\_v1\_PARTIAL surfaced honestly with named-SB attribution per brief's explicit prohibition.

## Merge Disposition

Merge report at /home/user/music-gen-instance/fork-33a2a8003c84/clone-0/merge\_report.md for root conductor pickup. Nine shadow-ledger rows queued for concat\_clone\_ledgers; zero cross-clone collisions expected under c32-v2 -clone-0 suffixes on infra families. Substantive M-EAR-1/real-label-training-v1 auto-suffixed per c37 extended-prefix note.

**Merge tasks:** 1. Post-merge concat of 9-event clone-0 shadow ledger. 2. Re-run PYTHON-PATH=. /usr/bin/python3 tests/test\_ear\_real\_label\_training\_v1.py post-merge; test\_20 should flip to PASS (23/23). 3. Register M-EAR-1/real-label-training-v1 in plan-of-record Milestones table if any peer-clone follow-up cycle intends to write further events under this family. 4. Carry three c39 handoff seeds into c39 research brief. 5. Keep workspace/harvest\_playlists.sh cycle-top retry going — SB1 is bound by corpus size, and only egress unblock lifts that bound.

## Cycle-39 Handoff (Priority Order)

1. **SB1 corpus expansion** — SB1 is bound by corpus size; the 43/80 partial-corpus caveat is the load-bearing constraint. Only egress unblock at \*.googlevideo.com (or operator hand-delivery of additional rated audio) lifts this.
2. **SB2 bootstrap-resample-count probe** — analytical study of the 10-resample threshold at N=43; may reveal that the c23-derived threshold is corpus-size-dependent.
3. **SB3 alternative-statistic candidate** — F1 saturates on singleton corpora by construction (c37 rubric §10 pre-registered). Candidate: within-artist repeat-corpus expansion (needs egress unblock + operator curation) OR F3 conditional- $\eta^2$  shrinkage (near-tie backup at c37 clone-1,  $\Delta = 0.020$ ) as a fallback statistic on singleton corpora with different degeneracy characteristics.
4. **\_manager/subagent-ledger-hygiene** — first-observed pattern this cycle: subagent lands substantive artefacts but fails to emit ledger events; auditor's post-hoc emission with agent="auditor" is the correct recovery path but unusual enough to codify a convention. Consider a \_manager/subagent-ledger-hygiene handoff for c39.

## Cumulative Progress

**M-EAR-1 arc** (post-c38 clone-0):

| Cycle       | Milestone                                                                                                   | Verdict                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| c22-c25     | Path A chassis chain                                                                                        | insufficient (three-audit chain; anti-patterns locked)                                                |
| c26         | _manager/M-EAR-1-path-B-commit<br>armed-harness-fixture-<br>reinforcement<br>M-EAR-1/real-label-training-v0 | committed; three SBs frozen                                                                           |
| c31         |                                                                                                             | FIXTURE_READY                                                                                         |
| c36         |                                                                                                             | EAR_v0_INSUFFICIENT (first real-label fire; SB3 statistic-degeneracy blocker surfaced)                |
| c37 clone-1 | _manager/ear-sb3-statistic-degeneracy-fallback-statistic                                                    | F1_ADOPTED (SB3 statistic-degeneracy blocker discharged)                                              |
| c38 (this)  | M-EAR-1/real-label-training-v1                                                                              | <b>EAR_v1_PARTIAL</b> (F1 lifted into leak_test; c26 Path B commitment discharged for the first time) |

**c26 Path B commitment discharged for the first time.** The armed-harness synthetic-fixture verification (c26) and the SB dry-run scaffolding (c31 clone-C) both anticipated this cycle; the real-label training pass now empirically fires and the three success bars evaluate honestly. EAR\_v1\_PARTIAL is neither a chassis failure nor a rubric failure — it is a corpus-size + corpus-shape (singleton-artist) finding.

**Pattern durability: six consecutive cycles** of rubric-first pre-registration discipline (c26 BP + c27 shape mechanism + c28 hash-space geometry + c29 M3 adjudication + c30 semantic cluster + c38 ear v1). Every cycle since c26 has committed a verdict rubric before analysis, with rubric SHAs embedded verbatim in verdict JSONs and mtime + git-log dual gates enforced. Zero after-the-fact rubric edits across the campaign.

**Fanout auto-suffix convention durable:** c32/c33 harness guard auto-suffixed all 9 c38 clone-0 events with -clone-0 (including the substantive M-EAR-1/real-label-training-v1/\* family, which is on the extended prefix list per c37 audit note). No ledger-concat conflicts anticipated at merge time.

**New pattern this cycle** (worth c39 codification): subagent-ledger-hygiene gap. Worker’s subagent landed substantive artefacts but failed to emit ledger events; auditor recovered via the role-authorised post-hoc emission path. First observed in the campaign; consider \_manager/subagent-ledger-hygiene handoff.

**Session-hygiene note:** worker-side hallucination early in the turn (fabricated verification results) was self-corrected within the same turn. Auditor verified actual on-disk state independently before acting. This is the pattern the “trust but verify” discipline exists for; documented here so future auditors know to disregard fabricated content if it appears in the transcript.

**c29 state-machine lemma** respected: peer sub-milestone; ledger topology stays a DAG.

**Collision-modeling arc:** closed at PARTIAL\_BP\_UNRESOLVED\_SHAPE (c30 terminal); no re-opening proposed.

**Rated audio egress:** still 403 at \*.googlevideo.com. Analytical + fixture-based work continues unblocked; M-EAR-1 real-label posture is now empirically anchored at EAR\_v1\_PARTIAL with

the corpus-size + singleton-corpus-shape findings the load-bearing constraints for any future v2/v3 cycle.

[END OUTPUT]